INCH-POUND
MIL-PRF-64266/9
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PERFORMANCE SPECIFICATION SHEET

# CONNECTORS, FIBER OPTIC, CIRCULAR, ALIGNMENT SLEEVE RETAINER (ASR), MULTIPLE REMOVABLE GENDERLESS TERMINI, SCREW THREADS, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring fiber optic connectors described herein shall consist of this specification sheet and MIL-PRF-64266.

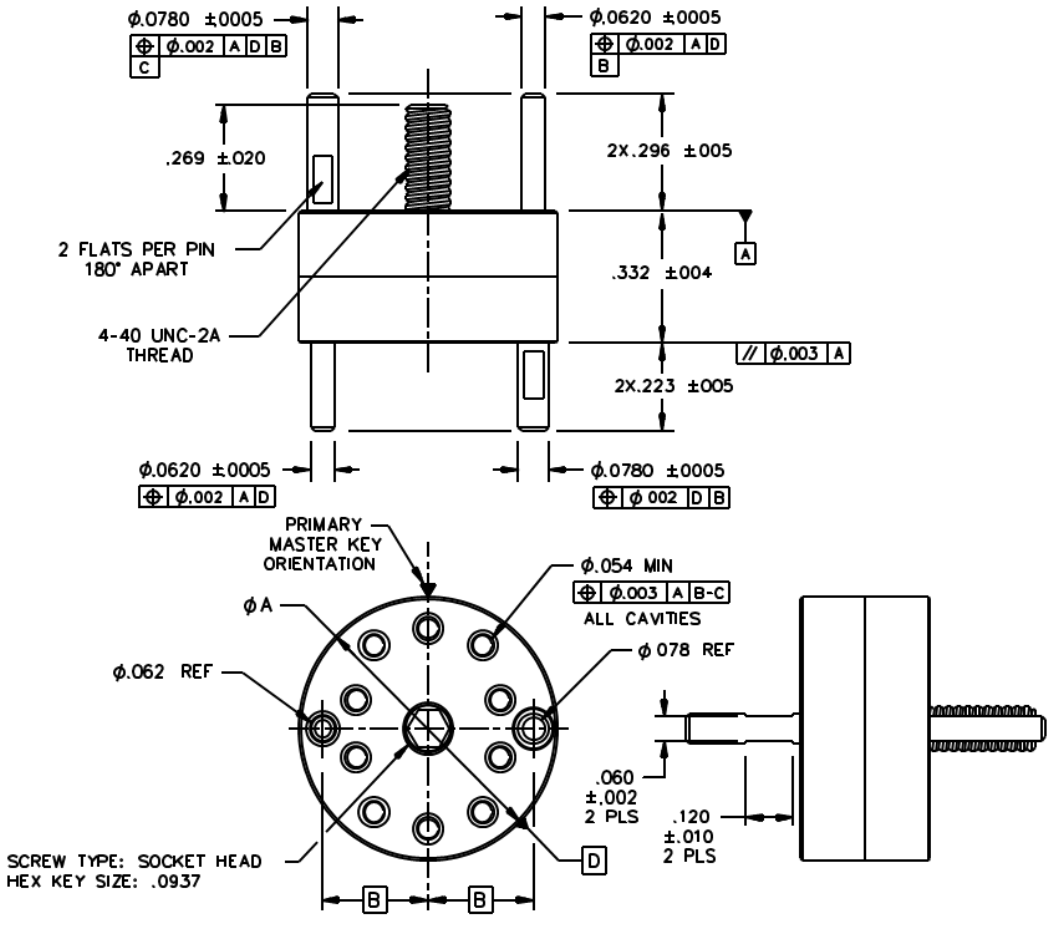



FIGURE 1. Alignment sleeve retainer (ASR).

AMSC N/A FSC 6060

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| Shell size | A dia +/- .002 | B basic (guidepins) |
|---|---|---|
| 11 | .410 (10.41) | .140 (3.56) |
| 13 | .486 (12.34) | .178 (4.52) |
| 15 | .664 (16.87) | .270 (6.86) |
| 23 | 1.068 (27.13) | .400 (10.16) |

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .0005 | 0.01 | .010 | 0.25 | .062 | 1.58 | .223 | 5.66 |
| .002 | 0.05 | .020 | 0.51 | .078 | 1.98 | .269 | 6.83 |
| .003 | 0.08 | .054 | 1.37 | .0937 | 2.38 | .296 | 7.52 |
| .005 | 0.13 | .060 | 1.52 | .120 | 3.05 | .332 | 4.43 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Dimensions apply to plated/finished part.
5. Alignment sleeve shall have an internal lead-in chamfer or radius in lieu of a sharp edge.
6. The cavity markings reflect the marking shown on the connector receptacle insert. The same insert marking is mirrored in the connector plug.

FIGURE 1. Alignment sleeve retainer (ASR) - continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1 herein and appendix B of MIL-PRF-64266, figures B-1 through B-6 for termini cavity insert position and arrangement dimensions. ASR shall be marked with the identification number of the insert cavity positions on the face opposite the protruding threads of the socket head screw.

Screw retention: Socket head screw shall be retained (captive) within the ASR.

Weight (maximum): Weight shall not exceed the values as specified in table I for the applicable shell size and material.

Table I. Alignment sleeve retainer weights.

| Shell size | Composite | | Aluminum | | Stainless steel | |
|---|---|---|---|---|---|---|
| | Ounces | Grams | Ounces | Grams | Ounces | Grams |
| 11 | 0.10 | 2.8 | 0.12 | 3.4 | 0.20 | 5.7 |
| 13 | 0.12 | 3.4 | 0.14 | 4.0 | 0.30 | 8.5 |
| 15 | 0.14 | 4.0 | 0.21 | 6.0 | 0.50 | 14.2 |
| 23 | 0.30 | 8.5 | 0.50 | 14.2 | 1.30 | 36.9 |

Counterpart connectors: Receptacle and plug connectors specified in MIL-PRF-64266/1, MIL-PRF-64266/2 and MIL-PRF-64266/3.



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Unless otherwise specified, the ASR shall be installed in the receptacle.

Termini: For use with MIL-PRF-29504/18, MIL-PRF-29504/19 and MIL-PRF-29504/20.

Alignment sleeves:
Material: Ceramic or equivalent.
Interface: Alignment sleeve shall have an internal lead-in chamfer or radius in lieu of a sharp edge.
Withdraw force: Alignment sleeve shall meet a withdraw force between 1.0 N to 2.5 N when tested for breakaway frictional force in alignment sleeves in accordance with TIA/EIA-455-158 using a gauge pin meeting the outside diameter (dimension CK), surface roughness and cylindrical requirements specified in figure 2.2.2 of TIA-604-10.

Guide pins: Use of a low strength thread locking compound on threaded guide pins is permissible providing the locking compound meets environmental requirements and does not migrate to the ferrule end faces of the fiber optic termini.

Marking:

PIN: The PIN shall be marked on the ASR circular surface (side) and packaging (see figure I):

Connector type: The side of the connector shall be marked with the word “RECEPTACLE” or the word “PLUG” depending upon whether the identification numbers of the insert cavity positions on the front face are for a connector receptacle or for a connector plug.

| M64266 | /9 | X | X | X | X |
|---|---|---|---|---|---|
| Specification number | Specification sheet | Shell size designator (see table II) | Class (see table III) | Insert arrangement number (see table II) | Connector type (see table IV) |

PIN Examples: M64266/9BA1P
M64266/9HC4R

TABLE II. PIN designators.

| Shell size | Shell size designator | Insert arrangement numbers |
|---|---|---|
| 11 | B | 1, 2, 3, or 4 |
| 13 | C | 1, 3 |
| 15 | D | 1, 2, 3, or 4 |
| 23 | H | 1, 3 |



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TABLE III. Class.

| Class | Material |
|---|---|
| A | Aluminum with anodized non-conductive plating (see 3.3.2). |
| B | Composite with anodized non-conductive plating (see 3.3.2). |
| C | CRES, passivated (see 3.3.2). |

TABLE IV. Connector type.

| Type | Description |
|---|---|
| R | Receptacle |
| P | Plug |

Installation tools, removal tool, and ASR assembly tool for alignment sleeve: As specified in appendix A of NAVSEA drawing 8283460. A copy of this document can be obtained at web site: https://fiberoptics.nswc.navy.mil.

For qualified products listing, conformance, and periodic inspections, dummy termini shall be used in all unused cavities.

Referenced documents. In addition to MIL-PRF-64266, this specification sheet references the following documents:

| | |
|---|---|
| MIL-PRF-29504/18 | MIL-PRF-64266/2 |
| MIL-PRF-29504/19 | MIL-PRF-64266/3 |
| MIL-PRF-29504/20 | NAVSEA drawing 8283460 |
| MIL-PRF-64266/1 | TIA/EIA-455-158 |
| | TIA-604-10 |

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6060-2011-016)

Review activities:
Navy - AS
Air Force - 13, 19, 93, 99
NASA - NA

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